MILITARY SPECIFICATION

MANUALS, TECHNICAL: MAINTENANCE/OVERHAUL INSTRUCTIONS and MAINTENANCE/OVERHAUL INSTRUCTIONS WITH ILLUSTRATED PARTS BREAKDOWN (For Automotive Equipment)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Coverage. This specification prescribes the requirements for preparation of technical manuals covering the maintenance and overhaul of automotive equipment, i.e., trucks, tractors, truck tractors, semi-trailers, forklifts, etc. A maintenance/overhaul instruction manual will usually be required for a complex equipment item and a maintenance/overhaul instructions with illustrated parts breakdown manual will usually be required for non-complex equipment; however, the type of manual to be prepared for any particular item of equipment will be determined by the procuring activity.

1.1.1 Maintenance/overhaul instructions manual. This type of manual shall be prepared when the complexity of the equipment will not permit complete and adequate maintenance and overhaul instructions and parts breakdown information to be furnished in a combined manual prepared in accordance with the requirements and limitations prescribed herein. (Complete and adequate information is essential; instructions shall not be limited for the sole purpose of meeting the requirements for a combined maintenance/overhaul instructions with illustrated parts breakdown manual, in lieu of a maintenance/overhaul instructions manual and a separate illustrated parts breakdown manual.)

1.1.2 Maintenance/overhaul instructions with illustrated parts breakdown manual. This type of manual shall be prepared when the equipment is relatively simple (non-complex), and complete and adequate maintenance and overhaul instructions and parts information can be furnished in a single manual prepared in accordance with the requirements and limitations prescribed herein. This type manual shall not exceed 36 pages, unless contractually authorized otherwise by the procuring activity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-M-38807	Manuals, Technical: Illustrated Parts Breakdown, Preparation of
MIL-M-38780	Manual, Technical: Nondestructive Inspection
MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

STANDARDS

Military

MIL-STD-681	Identification Coding and Application of Hookup and Lead Wire.

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications

AFLCR 65-42 Parts Kit

3. REQUIREMENTS

3.1 Preparation. The general manner of preparation shall be in accordance with MIL-M-38784 and MIL-M-38790.

3.1.1 Text content. Manuals shall include all data and procedures necessary for maintenance and overhaul of the equipment and its accessories. Parts information will be included in the form of illustrated parts breakdown when applicable.

3.1.1.1 Special processes. The manual shall include information on any special procedures or processes required under extreme temperature and humidity conditions within the limits established by the design specification.

3.1.1.2 Reference to components. Reference to components of the equipment shall be by type designation, standard nomenclature, location reference designation, or figure and index number, as applicable.

3.1.1.3 Color coding. Reference to color coding of chassis wiring shall normally be in accordance with MIL-STD-681. Chassis wiring in equipment which does not conform to the requirements of MIL-STD-681 shall be explained in the manual.

3.1.2 Illustrations. The requirements of MIL-M-38784 are applicable.

3.1.2.1 Diagrams. Diagrams of systems, such as fuel, oil and hydraulic, and internal wiring diagrams (schematics) shall be furnished when necessary and pertinent to the work described. When the circuitry is electronic rather than electrical, the schematics and other wiring diagrams shall show a reference symbol for each component included in the diagrams. In such cases, a table shall also be shown on, or adjacent to, the diagrams listing the components by basic noun nomenclature and model designation, and giving the blocks of reference symbol numbers assigned to each component.

3.1.2.2 Exploded views. Exploded views of the equipment shall be used for reference in disassembly and assembly instructions. Index numbers and a nomenclature list or legend, keyed to the index numbers, shall be used to identify each part. If the equipment is of such a nature that it cannot be adequately illustrated by a single exploded view, it may be exploded by subassemblies as separate views. In such cases, an exploded view showing the complete equipment exploded into its major assemblies shall be shown first. Parts which attach, and connect, the major assemblies together shall be shown on this illustration. These views, and those in a separate illustrated parts breakdown shall be the same, with the sequence of key index numbers in the order of disassembly.

3.1.2.3 Procedural illustrations. Essential illustrations depicting mechanical operations shall be included in pertinent sections, as necessary. It is not necessary to illustrate each step of an overhaul procedure, such as the removal of screws with an ordinary screwdriver, lifting off a cover after the screws have been removed, or the. Procedural illustrations should definitely supplement the text by clarifying procedures which are of a special nature or are not obvious.

3.1.2.1.4 Tool and test equipment illustrations. Standard tools and test equipment shall not be illustrated as such, nor shall self-evident or generally known uses be shown. Only uncommon or unusual uses, and connections for test purposes need be illustrated in appropriate sections if it is essential to do so to avoid misunderstanding as to proper procedures. Unusual operations in checking fits, clearances, backlash, etc., with feeler gages, torque wrenches, and micrometers shall be illustrated.

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3.2 Arrangement. Manuals shall be arranged in the following manner. Variation in sectionalization is permissable for certain types of manuals, dependent upon complexity of equipment. For non-complex items, material for two or more sections may be combined and some sections, not required (such as IPB), may be omitted. Section numbers and titles shall be changed accordingly. Manuals on simple items, ususally eight pages or less, may be divided into paragraphs insted of sections.

Front Matter
- Title Page
- List of Effective Pages
- Tables of Contents
- List of Illustrations
- List of Tables

Introduction

Section I	General Information
Section II	Special Tools and Test Equipment
Section III	Preparation for use
Section IV	In use Inspection, Maintenance, and Lubrication
Section V	Disassembly
Section VI	Cleaning
Section VII	Inspection, Repair, and Replacement
Section VIII	Assembly
Section IX	Accessories
Section X	Testing
Section XI	Table of Limits
Section XII	Illustrated Parts Breakdown
Section XIII	Difference Data Sheets

3.2.1 Front Matter.

3.2.1.1 Front matter shall be prepared in accordance with MIL-M-38784.

3.2.1.1.1 The introduction shall provide a brief explanation of the manual's purpose, scope, and arrangement including any relevant information which may increase the usability of the manual. The item(s) covered shall be identified by government designation. An illustration or exploded view shall be included.

3.2.1.1.2 If the manual will include an Illustrated Parts Breakdown and repair parts for the equipment are to be supplied in the form of kits, the introduction of the manual shall contain the following statement: "Some repair parts for the equipment covered by this manual are provided in the form of kits. The Illustrated Parts Breakdown section hereof identifies these parts. Activities shall replace all parts, regardless of condition, which are removed in the process of disassembly with all like parts furnished in the kit. Therefore, instructions for cleaning, inspecting, and reworking such used parts have been omitted. If any parts in the kit must be cleaned, inspected, or tested prior to installation, instructions for performing these requirements are included in the manual. Naturally, all defective parts are to be replaced; but a part unnecessary to be removed in the process of disassembly, shall not be removed solely for the purpose of replacement by a corresponding kitted part. Refer to AFLCR 65-42 regarding residue from kits and removed parts in this category."

3.2.1.1.3 If applicable, environmental control (clean room) information required for overhaul of the equipment shall be specified.

3.2.1.1.4 The applicable manuals (commercial or otherwise) associated with maintenance/overhaul will be referenced.

3.2.1.1.5 If warranty provisions are applicable, they will be stated.

3.2.2 Section I, General Information.

3.2.2.1 This section shall provide general information to be used by maintenance personnel to further their understanding of the equipment and its related systems. A brief description of leading particulars, and main features.

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3.2.2.2 If the manual covers more than one model, significant differences shall be explained. If "Difference Data Sheets" are contained in the manual, the introduction need only refer to them for explanation of differences.

3.2.3 Section II, Special Tools and Test Equipment. This section shall list and illustrate special tools and test equipment required for the disassembly, inspection, repair, assembly, and testing of the components and end item. Items shall be selected from the list of government approved tools and test equipment. When specified, the contractor's recommended tools and test equipment shall be shown. However, these recommendations shall be changed promptly, if necessary, to conform to the official government list as soon as such information is available. Acceptable alternate tools and test equipment will also be listed. Standard types of tools such as screw drivers, pliers, soldering irons, shall not be listed. Standard types of test equipment such as voltmeters, etc., shall be listed.

3.2.3.1 Special tools and test equipment list. Special tools and test equipment shall be listed in the following format:

SPECIAL TOOLS AND TEST EQUIPMENT LIST

Tool/Equipment Number	Figure No.	Nomenclature	Use and Application

a. Tool/Equipment Number-Tools shall be listed in sequence by part (tool) number. The part (tool) number shall agree with the drawing number. Test equipment shall follow the tool list and be listed in alpha-numberic sequence by type designation, using the AN type designation if assigned, or the commercial or manufacturer's designation if an AN type was not assigned. If an alternate item can be used, it will be listed immediately following the prime item and be identified as an alternate for the preceding item.

b. Figure No.-This column shall contain the number of the applicable illustration in which the tool/equipment is shown.

c. Nomenclature-The nomenclature shall agree with the drawing.

d. Use and Application-The purpose of each item shall be stated. Reference to the applicable paragraph in the text will suffice. If the manual covers more than one model, application to each model shall be indicated.

3.2.3.2 Illustrations of special tools and test equipment. Special tools and test equipment shall be illustrated in this section unless they are adequately illustrated in other sections of the manual. Standard items need not be illustrated.

3.2.3.3 Test equipment hook-up. If applicable, instructions shall be included for fabricating interconnecting cable assemblies, etc., required for bench test set-ups and not normally supplied with the equipment.

3.2.4 Section III, Preparation for Use. This section shall describe and illustrate in detail all maintenance required to prepare equipment for use, i.e., all adjustments and precautions to be observed.

3.2.5 Section IV, In use Inspection, Maintenance, and Lubrication. This section shall include complete instructions for inspecting, servicing, lubricating and adjusting assemblies by maintenance personnel.

3.2.5.1 Malfunction Analysis and Remedial Action. A chart of the most common malfunctions that are likely to occur during operation of the equipment (in sequence of probability of occurrence), a statement of probable cause(s), and a satisfactory remedy for each cause shall be included. The chart shall be in the format shown in figure 1.

3.2.5.2 Scheduled Inspection. The necessary inspection instructions and points to be checked shall be identified. Regular inspection periods shall be expressed in miles, hours of operation, or maximum period of time that may elapse between inspections if the prescribed mileage or hours of operation have not been reached. This information may be presented in chart form as shown in figure 2. Contractors

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shall determine the correct ratio of hours-time-mileage for the item of equipment concerned, taking into consideration the average distance driven for a typical piece of equipment for a given period of time. In addition to the usual inspection, which is limited to examination of the equipment when it is idle, the use of special purpose gauges, and recording the functional efficiency of various units and assemblies while in operation shall be referenced. Information on these items may be obtained, upon request, from the procuring activity.

3.2.5.3 Maintenance. Complete instructions for maintaining the equipment while in service shall be provided. Only operations which can be done with hand tools normally available to all mechanics and which do not require the facilities of overhaul organizations shall be discussed in this section.

3.2.5.4 Lubrication. The parts of the equipment which require lubrication, the amount of lubricant, the service specification number of the lubricant, and the mileage, or time, at which lubrication is required, shall be listed. This information shall be presented in chart form similar to figure 3. Lubrication data shall be given for temperatures ranging from minus 65 to plus 160 degrees Fahrenheit.

3.2.6 Section V, Disassembly. This section shall present, by text and illustrations, any special procedures preliminary to inspection, repair, and replacement of parts. These special procedures consist of removal from shipping container or equipment, dismantling into major assemblies, subassemblies, and component parts. The text shall contain statements keyed to the index numbers of the illustration, explaining special precautions to be observed and mentioning the special tools to be used to remove or replace certain parts or assemblies. Critical dimensions to be taken during disassemble should be specific.

3.2.6.1 Inspect, Repair, As Necessary. Data shall be included to enable repair activities to determine when components, assemblies, and subassemblies require maintenance or repair, or when they shall be completely overhauled. Processing instructions shall be included to cover inspection, the extent of disassembly necessary and the detailed mandatory parts replacement information applicable to time replacement and forced removal components.

3.2.7 Section VI, Cleaning. This section shall describe any special cleaning methods required. The type of cleaning shall be specified, such as using a liquid, brushing away dust, blowing out with compressed air; if the latter, air pressure required shall be indicated. Requirements for special ventilation or controlled environment should be indicated. Precautions to be observed shall be listed. Instructions for after cleaning relating to the preservation of metal parts and surface treatment (such as paint touch-up) shall be provided. Cleaning instructions in connection with nondestructive inspection methods shall be included. Special tools, jigs, fixtures, necessary during cleaning shall be indicated. General cleaning instructions shall be omitted.

3.2.8 Section VII, Inspection, Repair, and Replacement.

3.2.8.1 Inspection

3.2.8.1.1 This portion of the section shall include inspection instructions that may be classed as special. This means inspections that may be peculiar to a certain part of the equipment and for which maintenance personnel are likely to have little or no knowledge. Inspection instructions shall encompass failures, wear, damage, corrosion, leakage, aging, burning, malfunctioning, deformation and deterioration that can be expected to occur during normal service of the unit. They shall not include inspections that would only be applicable to detection of manufacturing defects. Each subassembly or the parts thereof, shall be treated as a separate unit in any discussion of inspection procedure. Each inspection check shall be described as the sequence of repair is required.

3.2.8.1.2 Definitions, and illustrations when necessary, of descriptive terms such as "scoring" and "galling" shall be included.

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3.2.8.1.3 All parts and subassemblies shall be listed that will normally require dimensional check for wear, damage, and deformation. The limiting dimensions of tolerances shall be such that inspected parts may be separated into (a) acceptable without further processing, (b) reparable, (c) rejected. The relationship of these dimensional limits to the assembly clearances specified in the Table of Limits, Section XI, shall be such that parts not classified as "rejected" may be assembled with normal adjustments, and may reasonably be expected to be within the replacement maximum limit of the Table of Limits.

3.2.8.1.4 All parts and subassemblies shall be listed that require special inspections by non-destructive methods, such as magnetic particle, ultrasonic, eddy current, x-ray, and liquid penetrants. When such inspections are necessary, the information required by that portion of MIL-M-38780 titled "Subsequent Sections" shall be provided.

3.2.8.1.5 Particular attention shall be given to all critical areas. A chart shall be used to summarize conditions for checking individual parts. This chart shall include material specifications for metals when pertinent to making repairs.

3.2.8.1.6 A concise discussion shall be given of the Table of Limits as it applies to this section. Definite information shall be given on the correct interpretation of the "Replacement Maximum" column in the Table of Limits.

3.2.8.2 *Repair and replacement.*

3.2.8.2.1 Comprehensive instructions shall be provided for necessary repair and replacement oporations to restore the equipment to a completely satisfactory condition. Text will be supplemented with illustrations as necessary to describe complex procedures. Repair methods shall be specified for the correction of all deficiencies and conditions considered reparable when inspected but shall not cover assembled replaced as units because they cannot be economically repaired or overhauled. Description of common procedures shall be omitted. Specific data shall be provided to adequately describe acceptable repairs.

3.2.8.2.2 The application and operation of special tools and test equipment shall be described as they are required in the repair and replacement operations.

3.2.9 Section VII, Assembly. This section shall give detailed step-by-step instructions for the assembly of component parts into major sub-assemblies to form a complete end item. Assembly instructions to include all pertinent assembly criteria such as clearances, backlash dimensions, special torque values, peculiar safety wiring, end play and similar data, shall be provided. Appropriate tests for operation of a sub-assembly prior to final assembly shall be provided, including safety precautions to be observed. Instructions for rigging, adjustment and alignment, when required, shall be included in logical sequence. Approved corrosion-preventive compounds, special lubricants, gasket pastes, etc., shall be specified. Any inspections, lubrications, etc., which must be performed prior to mating of subassemblies which can not be accomplished at a later point in the assembly will be identified. Instructions for required assembly lubrication shall be included. Maintenance check requirements shall be integrated into the procedure, as applicable. Assembly instructions, such as "Reverse the Disassembly Sequence", shall not be used. Reference to the applicable procedure for checkout of the item shall be made at the end of the assembly instructions.

3.2.10 Section IX, Accessories. This section is optional and shall be included in manuals covering relatively complex items upon which accessories are installed and overhaul instructions for such accessories are included in the end item manual.

3.2.10.1 Separate overhaul manuals will normally be procured for major accessories. This section shall accordingly state that for overhaul instructions on the major accessories not covered in this section, reference should be made to the applicable separate manuals.

3.2.10.2 This section shall contain applicable replacement, disassembly, special cleaning, inspection, repair, assembly, and test instructions on the minor accessories.

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3.2.11 Section X, Testing. This section shall contain essential instructions for aligning, adjusting, testing, and calibrating the equipment after reassembly. Test pointing, test values and acceptable tolerances shall be given for each test, as applicable. Remedial action shall also be described for failures of equipment to meet the necessary tests. This may be in the form of a trouble shooting table, if appropriate, showing trouble, probable causes, and remedy. As far as possible, the remedies shall specify adjustments which can be made without disassembling the equipment. If operating instructions are needed by overhaul personnel to test the equipment properly, such instructions shall be included to the extent necessary.

3.2.12 Section XI, Table of Limits. This section shall consist of introductory material, the Table of Limits, a miscellaneous table, and charts showing the limit points and lubrication system. The introductory material shall include definitions of unusual terms and abbreviations; also, statement to the effect that a minimum and maximum are set up as ideal limits and that measurements not exceeding the replacement maximum permit the part to be continued in service.

3.2.12.1 The Table of Limits shall include all clearances, backlashes, endplays, etc., arranged in numerical sequence by reference number. Reference numbers shall be so assigned that a numerical sequence for them will place the limits for particular areas of the equipment, or for subassemblies, into groups. Limits contained in the Table of Limits sections shall not be stated in the test of other sections. Reference in the text to these limits shall be made by citing the reference numbers used in the tables. The table shall be a cross reference to the chart on which the reference number is located. The table shall have the following format:

Ref No.	Chart No.	Description	Minimum	Maximum	Replacement Minimum/Maximum

3.2.12.2 The Table of Limits shall be supported by illustrations (charts) to show clearly all points of limits. Illustrations shall not show limit values but only index or reference numbers with leaders to the point of clearance. The illustrations shall also show lubrication information. Pressure oil, return oil, scavenge oil, propellants and pump fluids shall be differentiated by cross-hatching or other drafting means or by colors, with a legend explaining the coding.

3.2.12.3 The miscellaneous table shall include spring pressures and torque limits.

3.2.13 Section XII, Illustrated Parts Breakdown. This section shall be included only in manuals of overhaul instructions with illustrated parts breakdown for non-comples equipment (Refer to 1.1.2).

3.2.13.1 This section shall be generally in accordance with MIL-M-38807. Introductory information such as; explanation of symbols and abbreviations, definitions of applicable source and reparability codes, vendors' names and addresses, etc., will be included to the degree necessary to assure effective use of the manual.

3.2.13.2 An exploded view illustration and Group Assembly Parts List shall be all that is required for simple items. The format requirements of MIL-M-38807, are applicable.

3.2.13.3 Special tools and test equipment shall not be listed in this section since they will appear in the overhaul portion of the manual.

3.2.14 Section XIII, Difference Data Sheets. This section, when applicable, shall consist of Difference Data Sheets for additional models of the equipment. A separate Difference Data Sheet shall be prepared for each additional model to specify in the briefest practical form how overhaul and test procedures for that model differ from the procedures given for the basic model in preceding sections.

3.2.14.1 Format. The requirements of MIL-M-38784 are applicable. The Difference Data Sheet(s) for a particular model shall start on a right-hand page. Page numbers shall run consecutively throughout the section. They shall be added chronologically as they are issued.

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3.2.14.2 Content. Difference Data Sheets shall contain the following, as appropriate, to compress in one or two pages, and not more than four pages, the information on differences between the added model and the basic model covered in the preceding sections.

a. An external view of the model, if different than the one shown in Section I.

b. A table of leading particulars or a table of characteristic differences similar to that contained in Section I.

c. Illustrations which are essential for clarifying differences. Illustrations in preceding sections shall not be duplicated but referenced. Reference shall not be made to illustrations on other Difference Data Sheets.

d. Brief coverage of subjects specified for Sections II through IX by appropriate statements such as "Disassembly. Same as Model No. ----- except that -----." "Inspection. Same as Model No. -----." "Lubrication. None." Sentences shall be in short telegraphic style.

e. The last portion of the sheets for each model shall consist of a complete Group Assembly Parts List, as required by 3.2.13.2, or a list of differences arranged in Group Assembly Parts List format.

4. QUALITY ASSURANCE PROVISIONS

4.1 The quality assurance provisions shall be in accordance with MIL-M-38784.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784.

6. NOTES

6.1 Intended use. Technical manuals prepared in accordance with this specification are intended for use in the maintenance and overhaul of automotive equipment.

6.2 Supersession data. This specification supersedes that portion of MIL-M-38789A dated 1 July 1968 that pertains to automotive equipment and portions of MIL-M-26788B dated 1 November 1969 that apply to maintenance other than operator maintenance.

Custodian:

Air Force - 16

Review Activities:

Air Force -01, 10, 70

Preparing Activity

Air Force - 16

Project No. TMSC - F389

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TROUBLE	PROBABLE CAUSE	CHECKOUT PROCEDURE AND REMEDIAL ACTION
EXCESSIVE PUMP DISCHARGE PRESSURE.		
Pump discharge pressure is above 147 psi in all operating modes, engine speed above 2000 RPM.	1. Improperly adjusted, or defective main bypass pilot valve (V4OB-7). NOTE If condition persists after bypass pilot valve (V4OB-7) has been adjusted, check main bypass pilot valve system components for malfunction. 2. Defective main bypass valve (V4OB).	1.a. Readjust main bypass pilot valve. Refer to paragraph 5-21E. b. Replace main bypass pilot valve (V4OB-7) 2. Replace main bypass valve (V4OB).
Pump discharge pressure is above 58 psi with unit in "SINGLE POINT" operating mode, engine speed 2400 RPM, hose reel valves closed and pressure normal in other operating modes.	1. Improperly set, or defective bypass pilot valve (V4OB-1). 2. Bypass pilot valve (V4OB-2) closed. 3. Bypass pilot valve (V4OB-3) closed.	1.a. Readjust pilot valve to provide pressure of 58 psi maximum. b. If unable to reduce pressure, replace the pilot valve. 2. Repair or replace defective valve. 3. Repair or replace defective valve.
Pump discharge pressure is above 62 psi with unit in HI FLOW OVERRING operating mode, engine speed 2400 RPM, hose reel valves closed, and pressure normal in other operating modes.	1. Improperly adjusted or defective bypass pilot valve (V4OB-6.)	1. Readjust pilot valve to provide pressure of 62 psi maximum. 2. If unable to reduce pressure, replace the pilot valve.
Pump discharge pressure is above 130 psi but less than 147 psi when engine speed is 2400 RPM, Selector valve is in LOW or HI FLOW OVERRING position, and hose reel valves are closed	1. Observe nozzle inlet pressure gage (G1OC). If no pressure is indicated shutoff valve (V4OA) is closed and problem lies within the valve or its pilot system. a. Shutoff pilot valve (V4OA-3) not open. b. Shutoff pilot valve (V4OA-4) stuck closed c. Shutoff pilot valve (V4OA-5) stuck closed.	1.a. Refer to "Vent opens but shutoff valve (V4OA-3) does not actuate." b. Loosen control tubing at output side of valve. If fluid flow is not evident, repair or replace the valve. NOTE The valve is normally open. c. Loosen control tubing at output side of valve. If fluid flow is not evident, repair or replace the valve. NOTE The valve is normally open.

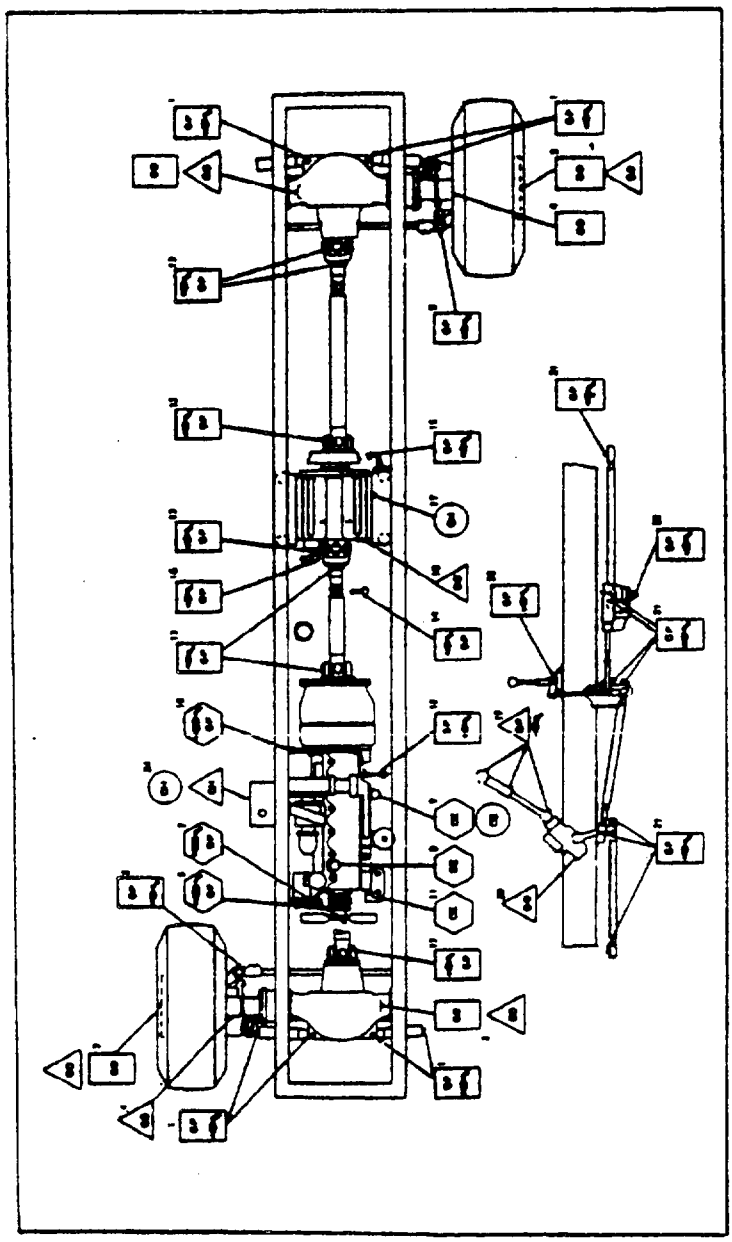
Figure 1 Trouble and Remedy Chart

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	INSPECTION INTERVAL	
	SCHEDULED	SAFETY
	3 months or 300 operating hours, whichever comes first	Annual, 12,000 miles, or 1200 whichever comes first
AIR SYSTEM		
Inspect all air lines for leaks.	X	
Check air tank and connections for leaks by using soap suds at connections. Tighten if necessay.	X	
Check air tank mounting. Tighten if loose.	X	
Check safety valve for leaks.	X	
Check safety valve connections.	X	
Check governor for leaks at air line connections.	X	
Check governor and tighten bolts.	X	
Check wiper arm and blade for tightness.	X	
Check wiper mounting studs.	X	
Check and tighten air line connections.	X	
Check all air line hoses for cracks and leaks.	X	
Check air line clamps and clips.	X	
Replace any worn or frayed air line.	X	
Check air compressor for mounting and belt tension.	X	
Clean oil supply lines.	X	
AXLES		
Check and tighten wheel driving flange bolts.	X	
Check axle thrust screws; tighten if necessary.		X
Check security of axle mounting to frame.		X
Check differentials for oil leaks.	X	
Check pinion nut tightness.	X	
Check differential and drive axle planetaries oil level.	X	
BATTERY		
Check gravity.	X	
Check terminals and cables for corrosion and tightness.	X	
BRAKES - VEHICLE		
Inspect linings for wear.		X
BRAKES - PARKING		
Check linkage for wear.	X	
Inspect linings for wear.		X
COOLING SYSTEM		
Check antifreeze.	X	

Figure 2 Example of Scheduled Inspection Chart.

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1. Steering cylinder
2. Differential bearing (see note 3)
3. Planetary housing (see note 4)
4. Universal joints and trunnion bearings (see note 5)
5. Tie rods
6. Fan pulley
7. Tachometer drive
8. Engine breather (see note 6)
9. Engine crankcase (see note 7)
10. Accelerator linkage
11. Lube oil cooler (see note 8)
12. Throttle linkage
13. Drive line bearings
14. Transmission shift rod
15. Parking brake ball crank
16. Speedometer adapter
17. Transmission sight gage (see note 9)
18. Transmission strainer (see note 10)
19. Steering column universal connector
20. Steering gear (see note 11)
21. Drag links
22. Steering shift lever
23. Steering ball crank
24. Hydraulic reservoir (see note 12)

Figure 2. Lubrication Chart (Sheet 1 of 2)

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LUBRICANT KEY

INTERVAL	SYMBOLS AND TYPES OF LUBRICANTS	TEMPERATURE RANGES AND LUBRICANT SPECIFICATION		
		32°F (0°C) AND ABOVE	0°F (-18°C) TO 32°F (0°C)	0°F (-18°C) TO -65°F (-54°C)
DAILY ○	OE LUBRICATING OIL, ENGINE	OE-30 MIL SPEC MIL-L-45199	OE-10 MIL SPEC MIL-L-45199	MIL SPEC MIL-L-45199 (See Note 13)
MONTHLY 100 HRS ⬡	HYDRAULIC STEERING OR OIL, HYDRAULIC	MIL SPEC MIL-L-2104 10W-30	MIL SPEC MIL-L-2104 10W-30	MIL SPEC MIL-L-2104 10W-30
QUARTERLY 300 HRS ▯	TRANSMISSION OR OIL, HYDRAULIC	TYPE C MIL-L-2104	TYPE C MIL-L-2104	TYPE C MIL-L-2104
ANNUALLY 1200 HRS △	OO OIL, GEAR	OE-90 MIL SPEC VV-L-765	OE-90 MIL SPEC VV-L-765	OE-90 MIL SPEC VV-L-765
	GP GREASE, GENERAL PURPOSE	MIL SPEC MIL-G-10924	MIL SPEC MIL-G-10924	MIL SPEC MIL-G-10924

APPLICATION SYMBOLS



OIL CAN
HAND
GREASE GUN
BRUSH

LUBRICATION NOTES

1. Clean the lubrication fittings with dry cleaning solvent, Federal Specification P-D-680. Dry before applying lubricant.

2. Intervals specified are for normal operation. The maintenance officer may adjust the intervals to compensate for unusual climatic or operational conditions.

3. Check the lubricant level quarterly or 300 operating hours whichever is first. Drain the lubricant while the differential is heated annually or 1200 hours and flush. Refill with new lubricant.

4. Check the lubricant level quarterly or 300 hours. Drain the planetary housing annually or 1200 operating hours. Flush and refill with new lubricant.

5. Drain the universal joint housing annually or 1200 hours. Flush and refill with new lubricant.

6. Clean the engine breather each month or 100 hours. Oil with OE lubricant.

7. Check oil level daily. Drain the crankcase every month or 100 hours. Refill with new OE lubricant.

8. Drain each time the engine crankcase is drained

9. Check oil level daily.

10. Drain the transmission annually or 1200 hours. Clean the strainer.

11. Drain and flush annually or 1200 hours. Refill with new lubricant.

12. Check the hydraulic oil level daily. Drain and refill with new oil annually or 1200 hours.

13. Dilute engine lubricating oil with Kerosene, Military Specification VV-K-211, for temperatures below 0°F (maximum dilution 25%).

Figure 3. Lubrication Chart (Sheet 2 of 2)

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